Feature Engineering & Nonlinearity

QLS–MiCM Workshop

Why This Matters

Real biological and clinical effects are **rarely linear**.

Examples:

- dose–response curves.
- saturation and thresholds.
- diminishing returns.
- nonlinear gene/protein interactions.

Key Idea: Add Nonlinear Features

We keep using a **linear model**, but expand the input features:

Polynomial Features

- x^2, x^3.
- interactions ($x_1 x_1, x_1^2, x_2, \ldots$).
- allows curvature and richer patterns.

This is still simple, interpretable, fast.

Workflow You'll Use

1. Load the drug response dataset.
2. Fit a **baseline linear model**.
3. Add polynomial features (degree=3).
4. Compare model performance.
5. Visualize:

- data vs linear fit.
- data vs polynomial fit.

What to Look For

- Do non-linear functions of x improve MAE?
- Does the polynomial model better follow the data trend?
- Is the relationship between top predictor + effect nonlinear?
- When does added flexibility risk overfitting?